

Darwix AI Voice Intelligence Platform

AI Engineer Assessment Submission

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Coverage: Questions 1-4 | Local working prototype | Synthetic demo data

GitHub repository

https://github.com/pankaj29-noi/Darwix_AI_Assignment

Video walkthrough

Not provided

Submission snapshot

- 34 automated tests passed; 0 failed.
- 6/6 RAG evaluation cases marked CORRECT by the executed rubric.
- Full real-time demo produced 6 signals and 5 evidence-backed nudges.
- Neutral and noisy false-positive scenarios produced 0 nudges.
- Latest local end-to-end latency: P50 1.178541 ms; P95 1.426871 ms over 7 chunks.

Important disclosure: the web interface is not a phone call. Cloud ASR quality/latency, LLM latency, and Filipino TTS quality are NOT MEASURED. No result is inferred or fabricated.

Q1 + Q2: Grounded voice agent and knowledge base

Q1 - Health-insurance lead qualification

- Conversation flow: greeting, consent, discovery, qualification, FAQ/objection, preliminary eligibility, callback/escalation, summary, and end.
- The voice agent calls the same KnowledgeBase.answer path used by the retrieval API; policy and objection facts are not embedded in the system prompt.
- Unsupported questions return a fixed safe fallback and can route to a human.
- The UI is a clearly labeled mock web session. A mock CRM summary and callback preference are stored.
- Executed scenarios cover cooperative, objection, incomplete, conflicting details, out-of-scope, and human assistance.

Q2 - Traceable RAG pipeline

- Input support: PDF, TXT, Markdown, HTML, and CSV.
- Pipeline: extraction -> cleaning -> normalization -> dedupe/near-dedupe -> PII redaction -> section-aware chunking -> metadata -> TF-IDF/FAISS -> reranking -> threshold -> cited answer or fallback.
- Records retain record_id, title, content, category, source, source_url, section, version, timestamps, PII, language, document_type, chunk_id, and hash.
- Six executed queries cover product, policy, qualification, FAQ, objection, and out-of-scope.

Executed retrieval results

Type	Top record	Score	Verdict
Product	kb_health_product_c001	0.4464	CORRECT
Policy	kb_health_policy_c001	0.4166	CORRECT
Qualification	kb_health_qualification_c001	0.2743	CORRECT
FAQ	kb_health_faq_c001	0.3710	CORRECT
Objection	kb_health_objection_c001	0.3078	CORRECT
Out of scope	safe fallback	0.0000	CORRECT

Q3 + Q4: Localization and live intelligence

Q3 - Philippines and Indonesia

- Philippines bancassurance supports English, Filipino/Tagalog, and natural Taglish, including premium, policy, beneficiary, rider, lapse, coverage, and bank referral.
- Indonesia multifinance supports formal and colloquial Bahasa plus finance terms such as cicilan, tenor, denda, DP, jatuh tempo, angsuran, and pembiayaan.
- At least three documented localization examples per market explain why wording is localized rather than literally translated.
- Fallback and escalation remain in the customer's language/register.
- The Javanese-influenced case reports expected meaning and a text-normalization preview. Actual acoustic accent recognition is NOT MEASURED.

Q4 - Streaming nudges before the replay ends

- Real-time-speed replay emits chunks through mock ASR, speaker-labeled transcript storage, signal detection, nudge generation, WebSocket delivery, and dashboard rendering.
- Signals include intent shift, compliance, frustration, buying intent, missed opportunity, callback, and payment difficulty.
- Nudges include evidence text, confidence, timestamp, and stage latency.
- Controls include confidence threshold, duplicate suppression, 45-second call-time cooldown, priority, topic grouping, expiry, and repetition limits.
- Full demo: 6 signals, 5 nudges. Neutral and noisy scenarios: 0 nudges each.

Measured local latency (milliseconds, n=7)

Stage	P50	P95
Mock ASR passthrough	0.001584	0.001921
Signal extraction	0.028541	0.039529
Nudge generation	0.005584	0.008488
Delivery/SQLite	1.130834	1.381591
End to end	1.178541	1.426871
LLM	NOT MEASURED	NOT MEASURED

These are in-process CPU/SQLite measurements with replay sleep excluded. They are not claims about cloud ASR, browser network delivery, or a hosted model.

Architecture, evidence, and production readiness

Architecture

Voice path: React web session -> ASRProvider -> VoiceOrchestrator -> KnowledgeBase.answer -> FAISS + SQLite -> extractive/live LLM generator -> TTSPProvider/browser speech.

Live path: real-time replay chunks -> ASRProvider -> SignalDetector -> NudgeEngine -> SQLite + WebSocket hub -> Live Intelligence dashboard.

Repository evidence

- README.md - setup, environment variables, usage, testing, and limitations.
- docs/ARCHITECTURE.md - component and data-flow diagrams.
- docs/Q1_VOICE_AGENT.md through docs/Q4_REALTIME.md - implementation decisions.
- results/ - executed RAG, multilingual, nudge, latency, and test JSON.
- transcripts/ - Q1, Philippines, Indonesia, and Q4 executed scenarios.
- recordings/MANIFEST.json - explicit synthetic-audio provenance and missing live-call disclosure.
- docs/PRODUCTION_PLAN.md - 10x scale, noisy audio, security, and provider improvements.

Known limitations

- No live telephony integration; the submitted interface is a web session.
- No credentialed cloud ASR or LLM run; those quality and latency claims remain NOT MEASURED.
- TF-IDF is the executed embedding path. The optional sentence-transformers path was not run.
- Speaker labels are provided by scripted scenarios, not acoustic diarization.
- No native Filipino system voice was available; synthetic audio must not be presented as a customer call.

Production improvements

Priorities: credentialed streaming ASR, tenant authentication, Postgres/pgvector, queues, encrypted PII, observability, provider fallbacks, native-speaker/compliance review, and load/noise evaluation at 10x concurrency.

Reviewer quick start

Install backend/requirements.txt and frontend dependencies, copy .env.example to .env, run FastAPI on port 8000 and Vite on port 5173, then follow docs/DEMO_SCRIPT.md.